LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Quick Revision: Magnetic Effects of Electric Current

Magnetic field & field lines

- A compass needle is a small bar magnet. Its north-pointing end is the north pole.
- Field lines go from N to S outside the magnet and from S to N inside, forming **closed curves**.
- Where lines are closer, the field is stronger. **Lines never cross**: a compass can't point two ways at once.
- Oersted (1820): a current-carrying wire deflects a compass needle, so **electric current produces a magnetic field**.

Field due to a current

Conductor	Field pattern	Strength increases with
Straight wire	Concentric circles around the wire	More current; closer to the wire
Circular loop	Circles near the wire; nearly straight lines at the centre	More current; more turns
Solenoid	Like a bar magnet; **uniform** (parallel lines) inside	More turns; more current; a soft iron core

- **Right-hand thumb rule:** thumb along the current; curled fingers show the field direction.
- **Electromagnet:** a soft iron core inside a solenoid. Magnetism can be switched on and off, and its strength changed.
- Example: current from east to west in an overhead wire. The field is north → south below the wire and south → north above it.

Force on a current-carrying conductor

- A conductor carrying current in a magnetic field experiences a force. The force is **greatest when the current is perpendicular to the field** and **zero when parallel**.
- Reverse the current or the field and the force reverses.
- **Fleming's left-hand rule:** forefinger = field, middle finger = current, thumb = force (motion), all at right angles.
- Used in electric motors, loudspeakers, microphones and measuring instruments. MRI scans use magnetic fields inside the body.

Domestic electric circuits

Wire	Insulation colour	Role
Live	Red	Carries current at 220 V (India)
Neutral	Black	Return path, at 0 V
Earth	Green	Safety: leaked current goes to earth, preventing shock

- Two circuits: **15 A** for high-power appliances (geyser, AC, heater) and **5 A** for bulbs and fans. All appliances are connected in parallel.
- **Short-circuit:** live touches neutral and the current shoots up. **Overloading:** too many appliances on one socket, or a voltage surge.
- **Fuse / MCB** breaks the circuit when the current exceeds the safe limit. A fuse is placed in the live wire.

Quick check

1. State Fleming's left-hand rule.

2. Why is the earth wire connected to the metal body of an appliance?